

SignalDesk — a news ranking engine that shows its work

A Telegram bot that reads 12 crypto and macro feeds, ranks every story with an auditable seven-factor score, and writes the briefing. Polymer Tech Expo 2026.

PROBLEM

Twelve newsfeeds produce a few hundred crypto and macro headlines a day — mostly listicles, price-prediction bait, and the same story told six times by six outlets. The signal is in there; the cost of finding it is forty minutes every morning.

The obvious fix — hand it to a language model and ask for a summary — fails in a specific way. **The model's selection is unauditable.** When it leads with the wrong story you cannot find out why, cannot reproduce it, and have nothing to correct. You have traded forty minutes of reading for a black box you must trust on a decision that matters.

SOLUTION

Split the job at its natural seam. **An algorithm decides what matters** — deterministic, inspectable, unit-tested. **A model writes it up** — it receives the already-ranked list and is told in its system prompt that selection is not its job: it cannot re-order, add stories, or introduce a number absent from the input.

- 0 • hard filters

clickbait, bare price ticks, macro noise — binary, so a listicle cannot out-vote its way back in
- 1 • dedupe

TF-IDF ∪ rare-entity overlap, grown to transitive closure
- 2 • seven-factor score

keyword ·222 recency ·222 source_quality ·220 category ·145 corroboration ·080 numeric ·056 asset ·056
- 3 • adaptive top-N

quality gate first, per-subject diversity cap, fixed count only as a floor
- 4 • LLM write-up → Telegram

daily digest, threshold alerts, live commands

/why 1 – "Curve Finance exploited for \$62m"

factor	raw	wt	adds
keyword	1.00	0.222	0.222
source_quality	0.95	0.220	0.209
recency	0.94	0.222	0.208
category	1.00	0.145	0.145
numeric	1.00	0.056	0.056
asset	0.50	0.056	0.028
source_count	0.00	0.080	0.000
TOTAL			0.867

Because every score carries its breakdown, `/why n` answers "why is this at the top" with arithmetic rather than a model's self-explanation. A wrong lead story traces to a factor weight — visible, testable, fixable — instead of to a sampling temperature.

USE OF AI

In the product: Moonshot Kimi K3 with xAI Grok 4.3 as fallback, via TokenRouter's OpenAI-compatible endpoint. The chain is deliberately cross-vendor: two models from one provider are a retry, not a fallback. The system prompt forbids any claim not grounded in the supplied titles and any number absent from the input. When every provider fails, the bot emits template output built from the top-ranked headlines verbatim — degraded, honest, still correctly ordered, because the ranking never depended on the model. **Building it:** Claude Code, as a pair programmer under a same-day deadline.

IMPACT

Forty minutes of triage becomes a thirty-second read, and — unlike a summariser — the output can be challenged. Marginal cost is one LLM call per digest: the twelve feeds, the Binance ticker and the Fear & Greed index are free and unauthenticated, so the data layer costs nothing to run or to scale. 68 offline tests cover the ranking path with no network and no API key.

REFLECTIONS

Making the score explain itself changed how I built it. `/why` began as a nice-to-have and became the debugging tool — **every real bug here was found by reading the factor table beside a ranking, not by a test.** A `\b(exploit)\b` regex could not match "exploited", so a \$62m protocol hack scored as untiered noise and ranked 7th.

The instructive one was self-inflicted. A fix for a missed duplicate *over*-merged on its first live run, because `ETF` passes a capitalisation test and read as a rare proper noun — collapsing unrelated stories that shared only "spot" and "ETF". The repair was a rarity threshold that **scales with batch size**: three mentions in eight headlines is a third of the feed; three in forty-one is genuinely rare. Both bugs were invisible in aggregate and obvious once the intermediate state was on screen — the argument for the whole design. **Next:** the weights are reasoned, not fitted — logging rankings against what readers actually open would make them measured.